Gotthold Eisenstein — Genaue Untersuchung der unendlichen Doppelproducte (1847)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the subject reviewers are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Ferdinand Gotthold Max Eisenstein of Berlin (1823–1852), who at twenty-four, because no one had ever proved such a thing, stopped his own theory to prove the general convergence lemma it needed, in an argument that still appears, in essence, in the textbooks — and whose memoir is met here in a partial rendering of its opening movement only, the rest still in German.

The author's own terms

Analytischer Modul M(u), his "analytical modulus" of a complex number u: its absolute value, $M(p + qi) = \sqrt{(p^2 + q^2)}$ — his period term, kept as he writes it. Σ′ (accented sums): sums over restricted index sets, the restriction, such as $M(u) \geq M(x)$, stated in place. *Doppelproduct* and *Doppelreihe*: the double product and double series over the two independent whole-number indices m and n; the lemma's "(α.)" and "(β.)" label its two coefficient systems.

The story

Gotthold Eisenstein was born in Berlin on 16 April 1823, the son of Johan Konstantin Eisenstein and Helene Pollack. Crelle recognised the genius in his

first papers and brought him to Alexander von Humboldt, who became his patron. Gauss wrote to Humboldt on 14 April 1846 of talents “such as nature bestows on only a few in each century.” He habilitated at Berlin in 1847.

In the Berlin street fighting of March 1848 he was dragged from a house from which shots had been fired, taken to the Spandau citadel, and mistreated. Released the next day, he never fully recovered, and the rumour that he was a “republican” dried up his funding. In late July 1852 he suffered a severe haemorrhage; Humboldt's Sicily funds came too late. He died of pulmonary tuberculosis in Berlin on 11 October 1852, aged twenty-nine, and the eighty-three-year-old Humboldt followed his coffin to the grave.

The work and what survives

The memoir, *Beiträge zur Theorie der elliptischen Functionen* VI, appeared in the *Mathematische Abhandlungen* (Berlin: G. Reimer, 1847) at volume pages 213–334 — *Crelle* 35 (1847), pp. 153–274 for the memoir as a whole (*Werke* I, pp. 357–478). Its subject is elliptic functions built as quotients of infinite double products. Section 1 excludes the degenerate real-ratio cases, keeping only the imaginary-ratio case, in which finitely many lattice points fall in any bounded region. It then proves the lemma his theory needs: the r-fold series $\Sigma\, 1/(m_1^{\,2} + \cdots + m_r^{\,2})^{\mu}$ converges when $\mu > r/2$, by counting lattice points in dyadic blocks. Section 2 proves the convergence of the product's logarithmic series.

The folder holds §§1–2 only — a partial slice in English draft: volume pages 213–230, the §§1–2 run at *Crelle* pp. 153–170, eighteen of the memoir's 122 printed pages. Sections 3 to the end (pp. 231–334) are absent, including the “third problem” announced at §2's close. The partial draft awaits review by a working mathematician and a reader of the period German. The machine-read text is good in the prose and treacherous in the displays; uncertain readings are bracketed, never silently repaired, and the load-bearing displays stand as flagged reconstructions from garbled machine text, uncertified pending page-image collation.

Its English fortunes

No English translation of this memoir was found after research, though its results are restated wherever elliptic functions are taught.

Restatement is not translation. Weil's *Elliptic Functions according to Eisenstein and Kronecker* and its Charollois–Sczech update rebuild his theory in modern language: exposition of his method, not a translation, and not to be cited as one. WorldCat and the union catalogues were not searched, nor Weil's full text for embedded passages; the finding stands at “none found after research” strength only, for the memoir whole and the partial slice alike.

What completing it would take

§§3 to the end — volume pages 231–334, about 104 printed pages — remain untranslated, roughly five times the partial slice in hand, and under the folder's slice ruling that work proceeds only on Robert's fresh word. First, the partial draft's load-bearing displays must be collated against the page images of the 1847 printing, ahead of the reviews by a working mathematician and a period-German reader it still awaits. The folder defers its map of the remainder to a notes file that is not in the folder; the map must be rebuilt from the 1847 volume's contents — open work — before the remainder can be scoped.

Sources

- Project folder: `~/workspace/rare-and-obscure-books/books/eisenstein-doppelproducte/eisenstein-doppelproducte.txt`
- PIN: `~/workspace/packages/boneyard-extension/research/PIN-eisenstein-doppelproducte.md`; life pinned in `~/workspace/packages/boneyard-extension/research/PIN-eisenstein-reciprocity.md`
- MacTutor History of Mathematics, Eisenstein biography: https://mathshistory.st-andrews.ac.uk/Biographies/Eisenstein/
- Charollois and Sczech, “Elliptic Functions according to Eisenstein and Kronecker: An Update”: https://hal.science/hal-03197278/document
- *Dictionary of Scientific Biography*, Eisenstein entry: https://un5ycdhmmurv21ygnwtdwuabdzg7ahkn9f0g.goodfellasdiner.com/DSB/Eisenstein.pdf